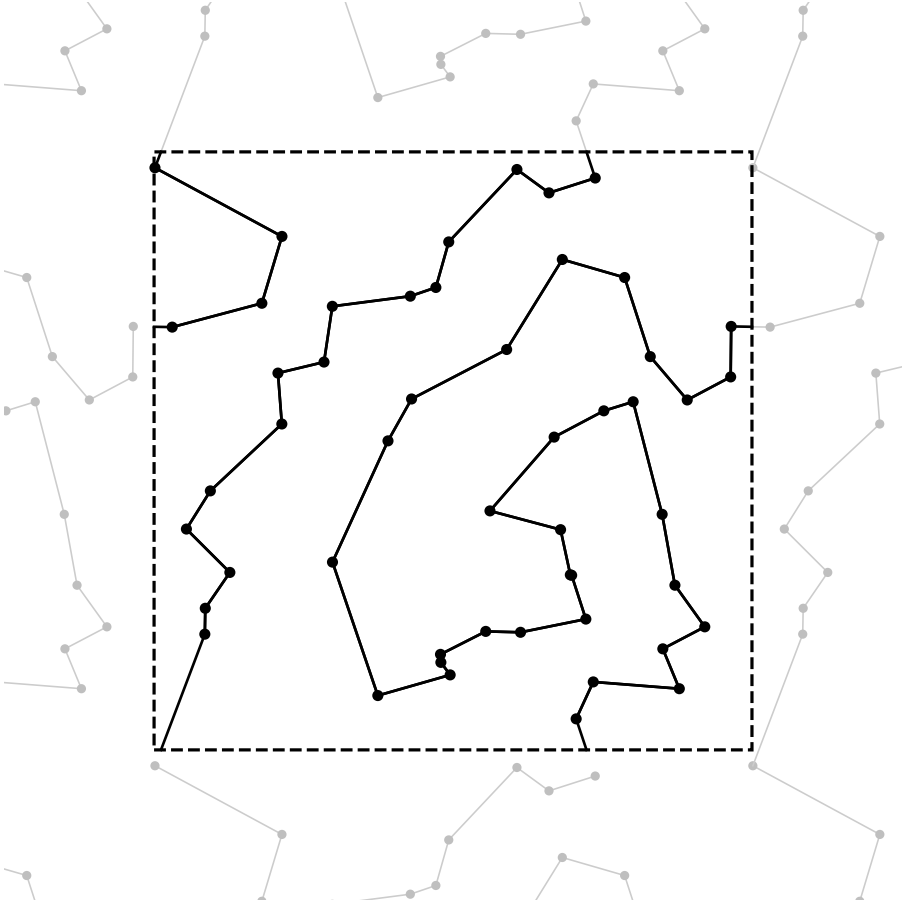
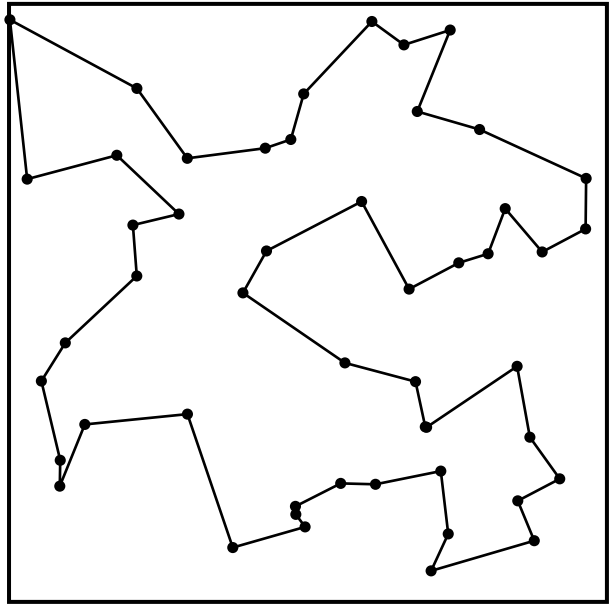
(a) Torus (periodic boundary): no boundary

$E[L_N^{\text{torus}}] \approx \beta\sqrt{NA}$



(b) Bounded region: boundary of length P

$E[L_N] \approx \beta\sqrt{NA} + \gamma_N P$